STAR: Spectral and Trend Adaptive Regime-Guided Forecasting for Multivariate Time Series

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***Abstract*—Existing multivariate time series forecasting models apply a fixed computational strategy irrespective of the statistical character of the input signal, leading to an inductive bias mismatch: spectral models degrade on aperiodic data, linear decomposition models fail to exploit cyclic structures, and Transformer- and state-space-based models achieve strong per-benchmark performance but generalize inconsistently across datasets and horizons. We propose STAR (Spectral and Trend Adaptive Regime-Guided forecasting), a regime-adaptive architecture that measures four interpretable properties of each input signal (spectral concentration, horizon ratio, cross-variate spectral co-occupancy, and linear trend strength) and routes predictions between complementary experts. A spectral-dynamic expert propagates frequency components with *horizon-adaptive exponential decay* and corrects residual non-periodic dynamics via delay-embedding cross-window attention. A trend decomposition expert applies moving-average decomposition with independent linear projection. A regime-conditioned Mixture-of-Experts router combines the two experts via soft gating, using spectral concentration, horizon ratio, cross-variate spectral co-occupancy, and trend strength as routing signals. Evaluated on ETTm1, ETTm2, ETTh1, ETTh2, and Exchange against eleven baselines including state-of-the-art models such as TimePro and iTransformer, STAR achieves the best or second-best average MSE on four of the five datasets and remains competitive on Exchange. The learned routing exhibits consistent regime-aware behavior, favoring the spectral expert on strongly periodic signals and the trend expert on trend-dominated or weakly periodic regimes.**

***Index Terms*—time series forecasting, mixture of experts, spectral forecasting, regime-adaptive routing, delay embedding, multivariate time series, spectral co-occupancy**

I. Introduction

Time series forecasting is a fundamental problem in machine learning, with applications spanning energy grid management, financial risk modeling, climate science, and clinical monitoring [1]. Despite sustained progress, accurately predicting multivariate sequences over long horizons remains challenging because real-world signals exhibit fundamentally different statistical regimes. Some display strong periodic structure with concentrated spectral energy, while others are trend-dominated with little exploitable cyclic behavior. Models that perform well under one regime often degrade under another due to inductive bias mismatch.

Existing approaches apply a fixed modeling paradigm regardless of input character. Spectral and frequency-domain models [2, 3] decompose signals into frequency components and propagate them forward, achieving strong results on periodic data but failing on aperiodic signals and extrapolating amplitudes at long horizons without any attenuation mechanism. Linear decomposition models [4, 5] demonstrate that trend-seasonal decomposition with linear projection can match complex architectures on trend-dominated signals, but have no mechanism to detect or exploit periodic structure when it is present. Transformer-based and state-space models [6-10] achieve strong benchmark performance through powerful sequence operators, but apply the same computation regardless of whether the input is strongly periodic, stochastic, or trend-dominated. Recent benchmark analyses confirm the consequence: model rankings change substantially across datasets and no single paradigm consistently dominates all evaluation settings [11].

We propose STAR (Spectral and Trend Adaptive Regime-Guided forecasting), which addresses this mismatch as follows. We measure four interpretable properties of each input signal (spectral concentration, horizon ratio, cross-variate spectral co-occupancy, and linear trend strength) and route predictions between complementary experts based on this regime descriptor. The cross-variate spectral co-occupancy term summarizes how broadly pairs of variates are jointly active across the spectrum, providing a lightweight cross-channel signal for the low-variate benchmarks evaluated here. As our first contribution, a spectral-dynamic expert combines horizon-adaptive exponential decay for frequency-domain propagation with a novel delay-embedding cross-window attention mechanism that corrects residual non-periodic dynamics. As our second contribution, a trend decomposition expert adapts the moving-average decomposition with independent linear projections of DLinear [4] as a complementary expert within a learned mixture, rather than as a standalone model. A regime-conditioned Mixture-of-Experts router combines the two experts via soft gating, using these interpretable regime descriptors as routing signals alongside a fixed horizon log-prior, with no dataset-specific configuration required. Experiments on ETTm1, ETTm2, ETTh1, ETTh2, and Exchange against eleven state-of-the-art baselines show that STAR achieves the best or second-best average MSE on four of the five datasets and the second-best average MAE on Exchange. Full results and routing analysis are reported in Section IV.

Our contributions are as follows:

- We propose a **regime-adaptive routing architecture** that selects between complementary forecasting experts based on input signal regime, addressing the inductive bias mismatch that causes benchmark-dependent performance variation in existing models.

- We introduce **Horizon-Adaptive Spectral Decay**, a theoretically motivated spectral propagation mechanism that prevents unrealistic amplitude extrapolation at long horizons by learning per-frequency decay rates conditioned on the forecast horizon.
- We propose an **Embedding-based Dynamic Module** (EDM), which uses delay-embedding cross-window attention to refine the spectral base forecast. EDM forms multi-scale delay embeddings from the concatenated history and provisional forecast, then lets forecast-window embeddings attend to history-window embeddings to estimate a residual correction, together with a **Spectral Centroid Alignment Loss** to stabilize joint training of the spectral and the dynamic branches.
- We design a **Regime-Conditioned MoE Router** that gates between the two experts using four interpretable signal descriptors (spectral concentration, horizon ratio, cross-variate spectral co-occupancy, and trend strength) together with a fixed horizon prior, and we demonstrate consistent regime-aware behavior through quantitative routing analysis.

II. Related Work

Transformer variants have been widely applied to time series forecasting, with early work focusing on attention modifications for long-range dependency modeling and computational efficiency [2, 12]. Subsequent approaches introduced stationarization [13], channel independence, and patch-based tokenization [7], improving performance across standard benchmarks. Crossformer [14] models cross-time and cross-variate dependencies through structured attention, while iTransformer [6] applies attention across variates rather than time steps to better capture inter-variable correlations. Despite these advances, Transformer-based models apply a fixed computational strategy independent of the input regime.

DLinear [4] demonstrated that simple trend-seasonal decomposition with linear projection can match complex architectures on several benchmarks. Lightweight MLP and mixing-based models [5, 10, 15] further showed that efficient parameterizations perform well on trend-dominated signals. State-space and Mamba-based methods [8, 9] offer scalable alternatives for long-range dependency modeling. In parallel, spectral approaches such as FEDformer [2], FITS [16], TimesNet [3], and FreTS [17] operate in the frequency domain and achieve strong performance on periodic data, but tend to degrade on aperiodic signals and may extrapolate amplitudes unrealistically at long horizons. Koopa [18] learns linear dynamics in latent space via Koopman operators, complementary to our delay-embedding-based dynamic refinement.

Recent evaluations show that model rankings vary substantially across datasets and horizons, with no single paradigm consistently dominating all settings [9, 11]. This variability highlights the importance of adapting model behavior to the underlying signal regime. STAR addresses this limitation by introducing regime-conditioned routing between complementary experts, enabling adaptive forecasting without dataset-specific configuration.

III. Methodology

Given a multivariate time series $\mathbf{X} \in \mathbb{R}^{B\times L\times N}$ observed over a lookback window of L timesteps across N variates, the forecasting objective is to predict $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times N}$ over a future horizon of H steps. We define the *horizon ratio* $\rho = H/L$, a scalar that explicitly conditions multiple internal components of STAR, i.e., spectral decay, the signal-adaptive fusion gate, and the routing log-prior, making the architecture horizon-aware at every stage rather than only at the output. Following standard practice [7], the primary forecasting computations operate *channel-wise* on $x_{cw} \in \mathbb{R}^{BN\times L}$, where B is the batch size, obtained by normalizing $\mathbf{X}$ with RevIN, *permuting to* $x_t \in \mathbb{R}^{B\times N\times L}$ and merging the batch and variate dimensions. The tensor x_t retains the variate dimension explicitly and is used by the trend expert and regime router where cross-variate structure is required. Cross-variate information enters through a separate regime descriptor $\boldsymbol{\phi}$, computed from the full input tensor, which guides the mixture-of-experts routing mechanism described in Section III-F.

Figure 1 illustrates the overall STAR architecture. The normalized input $x_{cw} \in \mathbb{R}^{BN\times L}$ feeds two parallel expert branches. The **base expert** combines a spectral propagation module with delay-embedding dynamic refinement and a signal-adaptive fusion gate, producing $\hat{y}_{base\_expert}$. The **trend expert** applies moving-average decomposition with independent linear projections, producing $\hat{y}_{trend}$. A **regime-conditioned MoE router** computes four interpretable regime features from the input and outputs soft routing weights γ_{base} and γ_{trend}, which combine the expert forecasts via weighted fusion. The mixture $\hat{y}_{mix}$ is denormalized by RevIN to produce the final forecast $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times N}$.

A. Reversible Instance Normalization

STAR applies Reversible Instance Normalization (RevIN [19]) as the first and last operation in the forward pass. Each variate is independently zero-centered and unit-variance normalized using statistics computed along the time axis of the lookback window. The per-variate mean μ_n and standard deviation σ_n are cached and applied in reverse after prediction, reducing the effect of scale and shift differences between the lookback and forecast windows on the loss and on the learned spectral decay rates.

B. Horizon-Adaptive Spectral Decay

Standard spectral forecasting methods, such as FEDformer [2], FITS [16], and TimesNet [3], propagate frequency components with amplitudes that are either fixed at the FFT values, capped by a learned scalar, or selected by a fixed energy budget. However, these approaches do not explicitly account for a well-established property of dynamic systems: the predictive information content of a frequency component

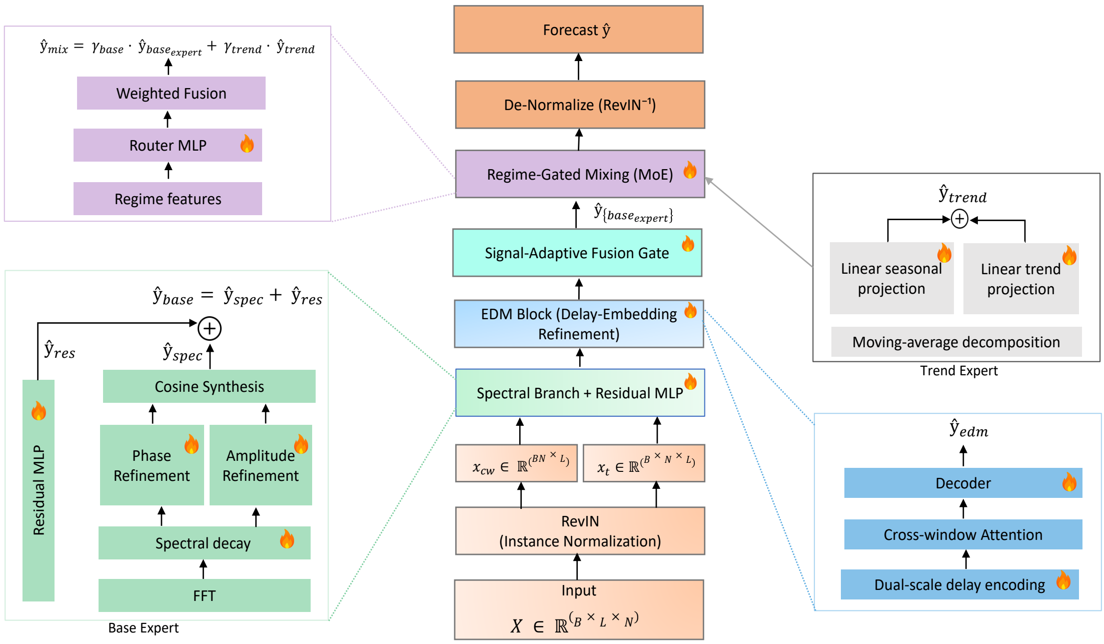

Fig. 1. **Overview of the STAR architecture.** The model routes predictions between a spectral-dynamic base expert and a trend decomposition expert via a regime-conditioned MoE router.

decays with forecast horizon, and this decay is more pronounced for higher frequencies. To address this, the spectral branch (Fig. 1, Base Expert) computes the one-sided real FFT of each channel x_{cw}, yielding $F = \lfloor L/2 \rfloor + 1$ complex coefficients with raw amplitudes $A_f^{raw} = |X_f|$ and phases $\varphi_f^{raw} = \angle X_f$ for $f = 0, \ldots, F-1$. The horizon ratio $\rho = H/L$ is appended as a constant scalar column, forming the conditioning tensor $\mathbf{inp} = [x_{cw} \,\|\, \rho] \in \mathbb{R}^{BN \times (L+1)}$, ensuring that downstream components are aware of both signal content and forecast horizon.

Learned frequency-dependent decay. A two-layer MLP with GELU activation, the *decay network* (MLP_β), maps $\mathbf{inp}$ to non-negative per-frequency decay rates:

$$\beta_f = \mathrm{softplus}(\mathrm{MLP}_\beta(\mathbf{inp})) \in \mathbb{R}_{\geq 0}^{BN \times F} \tag{1}$$

The $\mathrm{softplus}$ constraint ensures non-negativity: negative β_f would imply amplitude *amplification* with horizon, which lacks physical justification and empirically destabilizes long-horizon training. The output bias is initialized so that the decay factor is near unity at the start of training, allowing the branch to behave as a near-identity FFT extrapolator before learning frequency-specific attenuation. During training, high-frequency components with limited long-horizon predictability converge to larger β_f values, while low-frequency components corresponding to dominant periodicities converge to smaller β_f, receiving minimal attenuation.

Amplitude refinement and phase correction. A second MLP (MLP_r) applies a bounded multiplicative correction to the decayed amplitude, capturing small systematic deviations from pure exponential decay such as resonant amplification near dominant periods:

$$A_f^{final} = A_f^{raw} \cdot \exp(-\beta_f \cdot \rho) \cdot \big(1 + 0.1 \cdot \tanh(\mathrm{MLP}_r(\mathbf{inp})_f)\big) \tag{2}$$

A third MLP (MLP_φ) predicts bounded per-frequency phase corrections conditioned on the input signal and horizon ratio:

$$\varphi_f^{\text{final}} = \varphi_f^{\text{raw}} + 0.3 \cdot \tanh(\mathrm{MLP}_\varphi(\mathbf{inp})_f) \tag{3}$$

where the $\tanh$ nonlinearity bounds each phase correction. The grid frequencies $\omega_f = 2\pi f/L$ remain fixed; only the phase offsets are learned.

Spectral cosine synthesis. The spectral forecast is constructed by direct cosine summation rather than inverse FFT:

$$\hat{y}_{spec}(h) = \sum_{f=0}^{F-1} A_f^{final} \cdot \gamma_f \cdot \cos\Big(\omega_f h + \varphi_f^{final}\Big) \qquad h = 1, \ldots, H. \tag{4}$$

where γ_f are standard FFT normalization constants. Direct cosine synthesis allows each amplitude to be modulated freely by the decay and refinement terms without the conjugate-symmetry constraint that inverse FFT requires. Violating that constraint produces spectral artifacts that direct synthesis avoids entirely.

Spectral SNR (Signal-to-Noise Ratio) and centroid. The spectral branch produces two scalar side outputs consumed

TABLE I

MULTIVARIATE LONG-TERM FORECASTING RESULTS WITH PREDICTION LENGTHS $H \in \{96, 192, 336, 720\}$ AND FIXED LOOKBACK WINDOW L=96. THE *Avg* ROW REPORTS THE MEAN OVER ALL FOUR HORIZONS. SOTA BASELINE RESULTS ARE TAKEN FROM TIMEPRO [9]. RED: BEST RESULT. BLUE: SECOND BEST RESULT. LOWER IS BETTER FOR BOTH MSE AND MAE.

Dataset	Horizon	**STAR**		TimePro (2025)		S-Mamba (2025)		SOFTS (2024)		iTransformer (2024)		PatchTST (2023)		Crossformer (2023)		TiDE (2023)		TimesNet (2023)		DLinear (2023)		SCINet (2022)		Stationary (2022)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTm1	96	0.323	0.354	0.326	0.364	0.333	0.368	0.325	0.361	0.334	0.368	0.329	0.365	0.404	0.426	0.364	0.387	0.338	0.375	0.345	0.372	0.418	0.438	0.386	0.398
	192	0.367	0.377	0.367	0.383	0.376	0.390	0.375	0.389	0.377	0.391	0.380	0.394	0.450	0.451	0.398	0.404	0.374	0.387	0.380	0.389	0.439	0.450	0.459	0.444
	336	0.396	0.403	0.402	0.409	0.408	0.413	0.405	0.412	0.426	0.420	0.400	0.410	0.532	0.515	0.428	0.425	0.410	0.411	0.413	0.413	0.490	0.485	0.495	0.464
	720	0.455	0.437	0.469	0.446	0.475	0.448	0.466	0.447	0.491	0.459	0.459	0.453	0.666	0.589	0.487	0.461	0.478	0.450	0.474	0.453	0.595	0.550	0.585	0.516
	Avg	0.385	0.393	0.391	0.400	0.398	0.405	0.393	0.403	0.407	0.410	0.396	0.406	0.513	0.496	0.419	0.419	0.400	0.406	0.403	0.407	0.485	0.481	0.481	0.456
ETTm2	96	0.178	0.264	0.178	0.260	0.179	0.263	0.180	0.261	0.180	0.264	0.184	0.264	0.287	0.366	0.207	0.305	0.187	0.267	0.193	0.292	0.286	0.377	0.192	0.274
	192	0.242	0.302	0.242	0.303	0.250	0.309	0.246	0.306	0.250	0.309	0.246	0.306	0.414	0.492	0.290	0.364	0.249	0.309	0.284	0.362	0.399	0.445	0.280	0.339
	336	0.301	0.342	0.303	0.342	0.312	0.349	0.319	0.352	0.311	0.348	0.308	0.346	0.597	0.542	0.377	0.422	0.321	0.351	0.369	0.427	0.637	0.591	0.334	0.361
	720	0.394	0.396	0.400	0.399	0.411	0.406	0.405	0.401	0.412	0.407	0.409	0.402	1.730	1.042	0.558	0.524	0.408	0.403	0.554	0.522	0.960	0.735	0.417	0.413
	Avg	0.279	0.326	0.281	0.326	0.288	0.332	0.287	0.330	0.288	0.332	0.287	0.330	0.757	0.610	0.358	0.404	0.291	0.333	0.350	0.401	0.571	0.537	0.306	0.347
ETTh1	96	0.378	0.395	0.375	0.398	0.386	0.405	0.381	0.399	0.386	0.405	0.394	0.406	0.423	0.448	0.479	0.464	0.384	0.402	0.386	0.400	0.654	0.599	0.513	0.491
	192	0.423	0.423	0.427	0.429	0.443	0.437	0.435	0.431	0.441	0.436	0.436	0.435	0.471	0.474	0.525	0.492	0.436	0.429	0.437	0.432	0.719	0.631	0.534	0.504
	336	0.464	0.445	0.472	0.450	0.489	0.468	0.480	0.452	0.487	0.458	0.491	0.462	0.570	0.546	0.565	0.515	0.491	0.469	0.481	0.459	0.778	0.659	0.588	0.535
	720	0.510	0.489	0.476	0.474	0.502	0.489	0.499	0.488	0.503	0.491	0.487	0.479	0.653	0.621	0.594	0.558	0.521	0.500	0.519	0.516	0.836	0.699	0.643	0.616
	Avg	0.443	0.438	0.438	0.438	0.455	0.450	0.449	0.442	0.454	0.447	0.447	0.453	0.529	0.522	0.541	0.507	0.458	0.450	0.456	0.452	0.747	0.647	0.570	0.537
ETTh2	96	0.286	0.336	0.293	0.345	0.296	0.348	0.297	0.347	0.297	0.349	0.288	0.340	0.745	0.584	0.400	0.440	0.340	0.374	0.333	0.387	0.707	0.621	0.476	0.458
	192	0.364	0.389	0.367	0.394	0.376	0.396	0.373	0.394	0.380	0.400	0.376	0.395	0.877	0.656	0.528	0.509	0.402	0.414	0.477	0.476	0.860	0.689	0.512	0.493
	336	0.410	0.425	0.419	0.431	0.424	0.431	0.410	0.426	0.428	0.432	0.440	0.451	1.043	0.731	0.643	0.571	0.452	0.452	0.594	0.541	1.000	0.744	0.552	0.551
	720	0.421	0.439	0.427	0.445	0.426	0.444	0.411	0.433	0.427	0.445	0.445	0.436	1.104	0.763	0.874	0.679	0.462	0.468	0.831	0.657	1.249	0.838	0.562	0.560
	Avg	0.370	0.397	0.377	0.404	0.381	0.405	0.373	0.400	0.383	0.407	0.385	0.410	0.942	0.684	0.611	0.550	0.414	0.427	0.559	0.515	0.954	0.723	0.526	0.516
Exchange	96	0.084	0.204	0.085	0.204	0.086	0.207	0.091	0.209	0.086	0.206	0.088	0.205	0.256	0.367	0.094	0.218	0.107	0.234	0.088	0.218	0.267	0.396	0.111	0.237
	192	0.177	0.299	0.178	0.299	0.182	0.304	0.176	0.303	0.177	0.299	0.176	0.299	0.470	0.509	0.184	0.307	0.226	0.344	0.176	0.315	0.351	0.459	0.219	0.335
	336	0.326	0.412	0.328	0.414	0.332	0.418	0.329	0.416	0.331	0.417	0.301	0.397	1.268	0.883	0.349	0.431	0.367	0.448	0.313	0.427	1.324	0.853	0.421	0.476
	720	0.847	0.691	0.817	0.679	0.867	0.703	0.848	0.680	0.847	0.691	0.901	0.714	1.767	1.068	0.852	0.698	0.964	0.746	0.839	0.695	1.058	0.797	1.092	0.769
	Avg	0.359	0.402	0.352	0.399	0.367	0.408	0.361	0.402	0.360	0.403	0.367	0.404	0.940	0.707	0.370	0.413	0.416	0.443	0.354	0.414	0.750	0.626	0.461	0.454

by downstream components. Spectral SNR measures how concentrated the signal energy is in a few dominant frequency bins, with high values indicating strong periodicity and low values indicating stochastic or trend-dominated signals. The *spectral centroid* $\hat{\omega}_{spec}$ is the power-weighted mean angular frequency, characterizing the dominant timescale of the signal in a single scalar. Both are computed from the raw pre-decay amplitudes to reflect true signal properties. The SNR feeds the Signal-Adaptive Fusion Gate block in Fig. 1, and the centroid feeds the Spectral Centroid Alignment Loss block described in Section III-G.

The sum $\hat{y}_{\text{base}} = \hat{y}_{\text{spec}} + \hat{y}_{\text{res}} \in \mathbb{R}^{BN \times H}$, where $\hat{y}_{\text{res}}$ is the output of a residual MLP applied directly to x_{cw}, forms the provisional base forecast passed to the EDM block.

C. Delay-Embedding Dynamic Refinement

The spectral branch captures periodic, stationary structure effectively, but is fundamentally limited in its ability to model non-periodic dynamics, such as sudden regime shifts, transient events, and any structure that does not project cleanly onto the Fourier basis. The theoretical foundation for this component is Takens' embedding theorem, which establishes that for a smooth dynamic system with attractor of Hausdorff dimension d_A, the delay-coordinate map $\Phi : x(t) \mapsto [x(t), x(t-\delta), \ldots, x(t-(m-1)\delta)]$ is a diffeomorphic embedding of the attractor into $\mathbb{R}^m$ for almost every lag δ and any $m \geq 2d_A + 1$. This motivates representing each timestep by a vector of its recent lags and learning to query historical delay-embedded states that match the current forecast trajectory.

Dual-scale delay encoding. The EDM block operates on an extended sequence formed by concatenating the history window with the current forecast estimate: $e = [x_{cw} \,\|\, \hat{y}_{current}] \in \mathbb{R}^{BN \times (L+H)}$. The forecast estimate is *detached* from the computational graph before concatenation, preventing gradient flow from the dynamic refinement stage into the spectral branch. To capture dynamics at multiple temporal scales, two independent two-layer MLPs encode delay-coordinate representations at short and long lags δ_s and δ_l, mapping each position into a latent space of dimension M. The resulting representations are summed and layer-normalized, yielding $\boldsymbol{z} \in \mathbb{R}^{BN \times (L+H) \times M}$.

Asymmetric cross-window attention. The delay-embedded sequence $\boldsymbol{z} \in \mathbb{R}^{BN \times (L+H) \times M}$, output by the dual-scale delay encoder, represents the concatenated history and provisional forecast in a latent delay-coordinate space. It is split asymmetrically into forecast queries and history keys and values:

$$\mathbf{Q} = \boldsymbol{z}_{[L:,\,:]} \quad \mathbf{K} = \boldsymbol{z}_{[:L,\,:]} \quad \mathbf{V} = \boldsymbol{z}_{[1:L+1,\,:]} \tag{5}$$

where indices denote time-axis slicing. Queries correspond to the forecast window, while keys and values are drawn from the history window, with values shifted forward by one timestep to encode state-to-next-state transitions.

The cross-window attention matrix is

$$\boldsymbol{A}_{edm} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{M}}\right) \quad \boldsymbol{A}_{edm} \in \mathbb{R}^{BN \times H \times L} \tag{6}$$

The attended forecast representation is then

$$\boldsymbol{z}_{pred} = \boldsymbol{A}_{edm}\mathbf{V} \quad \boldsymbol{z}_{pred} \in \mathbb{R}^{BN \times H \times M} \tag{7}$$

Here, $\boldsymbol{z}_{pred}$ is a history-attended latent representation of the forecast window. A lightweight decoder maps this representation to a residual correction:

$$\hat{y}_{edm} = \hat{y}_{base} + c \cdot \text{Dec}(z_{pred}) \tag{8}$$

where $c \in \mathbb{R}$ is a learnable scalar that scales the EDM correction, allowing the spectral branch to stabilize first.

D. Bounded Signal-Adaptive Fusion Gate

After the EDM block produces $\hat{y}_{edm}$, a learned gate controls the contribution of the EDM correction to the base expert output, conditioned on spectral SNR and horizon ratio ρ through learned scalars u_{SNR}, u_{ρ}, and b:

$$\alpha = \alpha_f + (1 - 2\alpha_f) \cdot \sigma\Big(\text{softplus}(u_{SNR}) \cdot SNR + \text{softplus}(u_\rho) \cdot \rho + b\Big) \tag{9}$$

where α_f is a fixed floor and σ is the sigmoid function. The softplus constraint enforces non-negativity of the weights: a negative SNR weight would reduce α for high-SNR signals, increasing EDM influence precisely when spectral propagation is most reliable. The floor α_f prevents the gate from collapsing to exactly 0 or 1, which would suppress gradients to one branch, analogous to label smoothing.

The base expert output is:

$$\hat{y}_{base\_expert} = \hat{y}_{base} + (1 - \alpha)\,(\hat{y}_{edm} - \hat{y}_{base}) \tag{10}$$

E. Trend Decomposition Expert

The trend expert provides a complementary forecast grounded in classical time-series decomposition. A symmetric moving average with replicate padding decomposes each variate into a trend component and a seasonal-residual component. Two independent linear projections map each component from L to H timesteps and their outputs are summed. This design is closely related to DLinear [4], with the key difference that the trend expert is one option in a learned mixture rather than the sole model, and its weights are updated jointly with the spectral-EDM pathway through the shared loss. The expert produces $\hat{y}_{trend} \in \mathbb{R}^{BN \times H}$ and is entirely free of attention or nonlinear recurrence, making it both interpretable and computationally cheap.

F. Regime-Conditioned MoE Router

The MoE router (Fig. 1) requires a compact, interpretable descriptor of each input sample's temporal regime. We compute a four-dimensional feature vector $\boldsymbol{\phi} \in \mathbb{R}^{B \times 4}$ from quantities that are directly informative and computable from a single observed window.

Spectral regime descriptor. The first component, $\overline{SNR}$, is the mean spectral concentration ratio across variates. The second is the horizon ratio $\rho = H/L$, capturing how far the forecast extends beyond the lookback window. The third, $\bar{C}$, is the *mean pairwise spectral co-occupancy*, measuring the fraction of frequency bins over which pairs of variates carry power jointly:

$$\begin{aligned} C_{kj}(f) &= \frac{|X_k(f)|^2\,|X_j(f)|^2}{|X_k(f)|^2\,|X_j(f)|^2 + \varepsilon} \\ \bar{C} &= \frac{1}{N(N{-}1)} \sum_{k \neq j} \overline{C_{kj}(f)} \end{aligned} \tag{11}$$

where $X_k(f)$ denotes the one-sided rFFT coefficient of variate k, ε is a small numerical floor, and the overline denotes averaging over frequency bins. Computed from a single lookback window, $C_{kj}(f)$ is near one when both variates carry power at frequency f and near zero when either variate has negligible power at that frequency. Thus $\bar{C}$ summarizes the fraction of the spectrum over which a pair of variates is jointly active and should be interpreted as shared spectral co-occupancy rather than magnitude-squared coherence or phase coherence. For univariate settings ($N = 1$), $\bar{C} = 0$ by definition. The fourth component, $\bar{s}$, is the mean absolute OLS slope across variates and indicates the strength of linear trend in the input.

Prior-biased softmax routing. The regime descriptor $\boldsymbol{\phi}$ is mapped to two logits via a compact MLP, with a *fixed horizon log-prior* $\mathbf{p}(\rho) \in \mathbb{R}^2$ added before softmax:

$$(\gamma_{base},\, \gamma_{trend}) = \text{softmax}\big(\text{MLP}_{router}(\boldsymbol{\phi}) + \mathbf{p}(\rho)\big) \tag{12}$$

where $\mathbf{p}(\rho)$ favors the base expert at short horizons ($\rho \le 1$), remains base-favoring but attenuated for $1 < \rho \le 2$, is neutral for $2 < \rho \le 3.5$, and favors the trend expert at very long horizons ($\rho > 3.5$). The prior magnitude is kept small so that strong signal from $\boldsymbol{\phi}$ can override it, while still providing a sensible default when the descriptor is weakly informative. The routing weights lie on the probability simplex and are broadcast across variates, consistent with $\boldsymbol{\phi}$ aggregating cross-variate information.

The final prediction is the gated mixture:

$$\hat{y}_{mix} = \gamma_{base} \cdot \hat{y}_{base\_expert} + \gamma_{trend} \cdot \hat{y}_{trend} \tag{13}$$

followed by RevIN denormalization.

G. Composite Training Objective

The training loss combines three terms:

$$\mathcal{L} = \mathcal{L}_{TDT} + \lambda_{spec} \cdot \mathcal{L}_{spec} + w_{KS}(t) \cdot \mathcal{L}_{KS} \tag{14}$$

where λ_{spec} is a fixed auxiliary regularization coefficient held constant across all experiments.

Adaptive Temporal Derivative Loss. We introduce an adaptive temporal-derivative loss, $\mathcal{L}_{TDT}$, which combines point-wise reconstruction accuracy with first-difference shape alignment:

$$\mathcal{L}_{TDT} = \lambda \cdot \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + (1 - \lambda)\ \text{MAE}(\Delta\hat{\mathbf{Y}}, \Delta\mathbf{Y}) \tag{15}$$

where $\Delta\mathbf{Y}_{:,t,:} = \mathbf{Y}_{:,t,:} - \mathbf{Y}_{:,t-1,:}$ denotes the first difference, and λ is the fraction of time–variate pairs in the current batch whose predicted and target first differences have opposite signs, computed without gradient. When directional errors are frequent, λ is large and the MSE term dominates. As directional agreement improves, the derivative term receives higher weight and encourages local shape alignment. This results in a batch-adaptive weighting mechanism without an explicit schedule hyperparameter.

Symmetric Spectral Divergence. $\mathcal{L}_{\text{spec}}$ is the symmetric Kullback–Leibler (KL) divergence between the normalized forecast-horizon power spectra of the prediction and the target. The KL terms are summed over frequency bins and then averaged over batch elements and variates:

$$\mathcal{L}_{spec} = \frac{1}{2}\Big(D_{KL}(P_y \,\|\, P_{\hat{y}}) + D_{KL}(P_{\hat{y}} \,\|\, P_y)\Big) \tag{16}$$

where P_y and $P_{\hat{y}}$ are the normalized power spectra of the target and prediction, computed as squared FFT magnitudes normalized to sum to one over the frequency dimension. This formulation penalizes divergence in both directions, avoiding the instability of one-sided KL when near-zero mass is assigned to true spectral components. This term penalizes forecasts whose frequency-energy distribution differs from the target, even when their point-wise error is small. Thus, $\mathcal{L}_{\mathrm{spec}}$ complements $\mathcal{L}_{TDT}$ by encouraging the prediction to preserve the target's spectral structure.

Spectral Centroid Alignment Loss.

A subtle failure mode in two-branch architectures arises when the spectral and dynamic branches operate at incompatible dominant timescales, where the spectral branch may capture low-frequency periodicity while the EDM attention focuses on short-range transitions, leading to interference rather than complementary refinement. The Spectral Centroid Alignment Loss addresses this directly. The spectral centroid $\hat{\omega}_{spec}$ characterizes the signal's dominant timescale; the EDM centroid $\hat{\omega}_{edm}$ is defined as the power-weighted mean frequency of the attention trace, obtained by averaging the last EDM block's attention matrix across query positions and applying the FFT. The loss aims to minimize the squared distance between the two:

$$\mathcal{L}_{KS} = \mathrm{MSE}(\hat{\omega}_{spec}, \hat{\omega}_{edm}) \tag{17}$$

We use $\mathcal{L}_{KS}$ as shorthand for this spectral-centroid alignment loss, which is distinct from the classical Kolmogorov–Smirnov statistic. The loss weight is linearly ramped during training, allowing the branches to develop independently before alignment is enforced.

IV. Experiments

A. Experimental Setup

Datasets and baselines. We evaluate on five standard long-term forecasting benchmarks: ETTh1, ETTh2 (hourly electricity transformer temperature, N=7), ETTm1, ETTm2 (15-minute, N=7), and Exchange (N=8, daily currency exchange rates from eight countries), where N denotes the number of variates. Following the unified evaluation protocol of iTransformer [6] and TimePro [9], the lookback window is fixed at L=96 and prediction horizons are $H \in \{96, 192, 336, 720\}$. Data splits follow the chronological partitions of TimePro [9]: ETT datasets use (8,545/2,881/2,881) time points for hourly variants and (34,465/11,521/11,521) for 15-minute variants; Exchange uses (5,120/665/1,422) (train/val/test). We compare against 11 SOTA baselines including TimePro [9], S-Mamba [8], SOFTS [10], iTransformer [6], PatchTST [7], Crossformer [14], TiDE [5], TimesNet [3], DLinear [4], SCINet [20], and Stationary [13]. Baseline results are taken from TimePro [9] (Table I).

Implementation. All experiments are conducted on two NVIDIA RTX 6000 Ada Generation GPUs and implemented in PyTorch. We report MSE and MAE as evaluation metrics, where lower is better. STAR is trained with the AdamW optimizer, learning rate 10^{-4}, weight decay 10^{-3}, batch size 16, 5-epoch linear warmup followed by cosine decay over 295 epochs, and early stopping with patience 50. The EDM block uses delay parameters δ_s=2, δ_l=4, embedding dimension M=64, and a single refinement block. Loss weights are λ_{spec}=0.01 and w_{KS}=0.02 with a linear ramp from epoch 20 to 60; the α_f floor is 0.05.

B. Main Results

Table I reports the per-horizon and averaged results across all five datasets. STAR demonstrates consistently strong performance, achieving the best or second-best average MSE on four datasets, ranking first on ETTm1, ETTm2, and ETTh2, and second on ETTh1. On ETTh1, STAR further achieves the best average MAE, indicating strong alignment with target trajectories despite slightly higher MSE at long horizons. On periodic datasets where spectral structure is prominent (ETTm1, ETTm2), STAR achieves the best performance across nearly all horizons, tying with TimePro only at ETTm1 H=192. On ETTh2, STAR achieves the best average MSE (0.370), outperforming SOFTS (0.373) and TimePro (0.377). On ETTh1, STAR achieves the best results at H=192 and H=336, and the best average MAE, while ranking second overall by MSE due to reduced performance at H=720. On Exchange, STAR remains competitive, achieving the best result at H=96 and strong performance at shorter horizons; although it ranks third overall by average MSE, it attains the second-best average MAE, as performance degrades at longer horizons where weak periodic structure limits the effectiveness of spectral extrapolation.

Efficiency. Table V reports STAR's trainable parameters, multiply–accumulate operations (MACs), memory, and per-batch latency across horizons, profiled on ETTh1 (N=7, L=96, batch size 16) on a single RTX 6000 Ada GPU. A MAC consists of one multiplication and one accumulation, so the MAC count measures the arithmetic cost of a single forward pass. STAR uses 52.7K–194.3K parameters and runs in under 1.5 ms per batch at every horizon. The four ETT datasets share the same efficiency profile. In terms of model size, STAR uses far fewer parameters than typical Transformer-based long-term forecasters, which require millions of parameters (e.g., 13–21M for Transformer, Informer, Autoformer, and FEDformer [4]). We do not compare MACs, memory, or runtime across papers, as these quantities depend on batch size, hardware, implementation, and the number of variates; we therefore report them for STAR alone. A fully matched baseline efficiency study, profiling the baselines on identical hardware, is left for future work.

C. Reproducibility Analysis

Table II reports mean and standard deviation of MSE and MAE across five random seeds. Standard deviations remain at or below 0.007 for most reported results, with the largest variance at ETTh1 H=720 (MSE std 0.015, MAE std 0.008).

This elevated variance at the longest horizon is consistent with the mixed-regime routing behavior on ETTh1 documented in Section IV-D: when the router assigns borderline gate weights, small weight changes across seeds can shift the balance between experts and propagate to forecast error. ETTm2 is the most stable dataset with avg MSE std

TABLE II
REPRODUCIBILITY ANALYSIS: MEAN $\pm$ STD OF MSE AND MAE OVER FIVE RANDOM SEEDS, $L = 96$

Dataset	Metric	96	192	336	720	Avg
ETTh1	MSE	0.380 ± 0.004	0.423 ± 0.003	0.464 ± 0.002	0.487 ± 0.015	0.438 ± 0.003
	MAE	0.395 ± 0.002	0.423 ± 0.001	0.444 ± 0.001	0.476 ± 0.008	0.435 ± 0.002
ETTh2	MSE	0.287 ± 0.000	0.368 ± 0.003	0.416 ± 0.004	0.420 ± 0.002	0.373 ± 0.002
	MAE	0.337 ± 0.001	0.392 ± 0.003	0.427 ± 0.002	0.440 ± 0.001	0.399 ± 0.001
ETTm1	MSE	0.322 ± 0.002	0.365 ± 0.003	0.395 ± 0.007	0.456 ± 0.004	0.385 ± 0.003
	MAE	0.355 ± 0.001	0.379 ± 0.001	0.401 ± 0.001	0.436 ± 0.002	0.393 ± 0.001
ETTm2	MSE	0.177 ± 0.001	0.242 ± 0.001	0.301 ± 0.001	0.397 ± 0.002	0.279 ± 0.001
	MAE	0.263 ± 0.001	0.303 ± 0.001	0.341 ± 0.001	0.398 ± 0.002	0.326 ± 0.000
Exchange	MSE	0.085 ± 0.000	0.178 ± 0.001	0.327 ± 0.002	0.841 ± 0.003	0.358 ± 0.002
	MAE	0.205 ± 0.000	0.300 ± 0.001	0.413 ± 0.001	0.689 ± 0.001	0.402 ± 0.001

TABLE III
ROUTING GATE WEIGHTS AND REGIME FEATURES AVERAGED OVER FIVE RANDOM SEEDS ($\gamma_{\text{trend}} = 1 - \gamma_{\text{base}}$)

Dataset	γ_{base}	$\overline{SNR}$	$\bar{C}$
ETTh1	0.363 ± 0.032	0.766	0.980
ETTh2	0.424 ± 0.035	0.637	0.955
ETTm1	0.753 ± 0.055	0.852	0.979
ETTm2	0.621 ± 0.025	0.693	0.857

$= 0.001$. The overall average MSE across all five datasets is 0.367 ± 0.001, confirming that the reported results are robust to initialization rather than dependent on a favorable seed. The main-results seed is broadly consistent with the five-seed distributions reported in Table II, while the five-seed summary provides the more reliable estimate of initialization stability.

D. Analysis: Routing behavior

Table III reports routing diagnostics averaged across five seeds. Values above 0.5 indicate preference for the spectral-dynamic base expert, while values below 0.5 indicate preference for the trend expert. Although the variability of γ_{base} for ETTh2 and ETTm2 places these datasets close to the 0.5 boundary, the mean routing behavior remains consistent with the expected regime differences.

In the Spectral-dominant case (ETTm1, ETTm2), both 15-minute ETT datasets assign majority weight to the spectral-dynamic expert ($\gamma_{\text{base}} = 0.753$ and 0.621), consistent with higher spectral concentration ($\overline{SNR} \geq 0.69$) and broad shared spectral support ($\bar{C} \geq 0.85$). In the case of trend-dominant (ETTh1, ETTh2), both hourly ETT datasets route toward the trend expert ($\gamma_{\text{trend}} = 0.637$ and 0.576), despite moderate-to-high SNR values. This indicates that the router integrates multiple regime signals. For example, ETTh1 has higher SNR (0.766) than ETTm2 (0.693), yet routes toward the trend expert, while ETTm2 routes toward the spectral expert, showing that factors beyond SNR govern the decision. The ETTm–ETTh split reveals an interpretable pattern: the 15-minute resolution of ETTm datasets preserves intra-day periodic structure exploited by the spectral expert, whereas the hourly ETTh datasets favor trend dynamics over high-frequency components.

E. Ablation Study

Table IV presents a component-wise ablation study where each row removes exactly one component from the full STAR model at seed 42 with L=96.

Normalization. Removing RevIN causes by far the largest degradation of any variant: overall MSE increases by 0.055, with the largest per-dataset effect on ETTh2 (+0.163 MSE) and ETTm2 (+0.044 MSE), confirming that per-variate instance normalization is critical across all signal regimes.

Spectral branch components. Removing the full spectral branch degrades overall MSE by 0.005, comparable to removing spectral decay alone (+0.005), while individual sub-component removals (amplitude refinement, phase correction) cause smaller effects (+0.003–0.004), indicating that the spectral sub-components are partially redundant and jointly contribute to the branch output.

EDM blocks. Removing EDM blocks causes the second-largest architectural degradation (+0.007 overall MSE), with larger per-dataset effects on ETTh1 (+0.013) and ETTh2 (+0.009) than on ETTm1 (+0.005) and ETTm2 (+0.002), consistent with EDM correcting residual dynamics that the spectral branch leaves unmodeled.

Expert routing. The routing ablations reveal a clear asymmetric pattern. Removing the trend expert degrades ETTh2 by 0.019 MSE but ETTh1 by only 0.002, confirming that the trend expert is critical on trend-dominated signals. Removing the learned router degrades ETTh1 by 0.012 but ETTh2 by only 0.003, showing that routing provides the largest benefit on signals where the spectral expert is appropriate and must be reliably selected. This asymmetry directly validates the regime-routing motivation of STAR.

Loss functions. Removing $\mathcal{L}_{KS}$ and $\mathcal{L}_{TDT}$ degrades overall MSE by +0.003 and +0.002, respectively, indicating that both terms contribute meaningfully to performance. In contrast, removing $\mathcal{L}_{spec}$ leaves overall MSE essentially unchanged (-0.001), a difference on the order of the seed-to-seed variation reported in Table II. This behavior is consistent with the role of $\mathcal{L}_{spec}$ as a low-weight (λ_{spec}=0.01) regularizer that targets the frequency-domain structure of the forecast rather than directly optimizing point-wise error. We therefore retain $\mathcal{L}_{spec}$ as a low-weight spectral-structure regularizer rather than a point-accuracy objective. The signal-adaptive fusion gate has only a minor effect (+0.001 overall), suggesting that the internal spectral–EDM balance remains relatively stable even without explicit gating.

V. CONCLUSION

We presented STAR, a multivariate time series forecasting model that measures four interpretable signal properties and routes predictions between complementary experts without dataset-specific configuration. Across five benchmarks against eleven baselines, STAR achieves the best or second-best average MSE on four of five datasets. Routing analysis confirms that assignments align with signal structure: 15-minute ETT datasets route to the spectral-dynamic expert while hourly ETT datasets route toward the trend expert, a distinction that

TABLE IV

COMPONENT ABLATION: AVERAGE MSE / MAE OVER $H \in \{96, 192, 336, 720\}$, L=96. EACH ROW REMOVES EXACTLY ONE COMPONENT FROM THE FULL STAR MODEL. LOWER IS BETTER. **BOLD**: FULL MODEL. ↑ INDICATES DEGRADATION; ↓ INDICATES IMPROVEMENT.

Variant	ETTh1		ETTh2		ETTm1		ETTm2		Exchange	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o RevIN	0.453 ↑	0.448 ↑	0.533 ↑	0.482 ↑	0.411 ↑	0.415 ↑	0.323 ↑	0.366 ↑	0.390 ↑	0.443 ↑
w/o Spectral decay	0.455 ↑	0.445 ↑	0.380 ↑	0.402 ↑	0.380 ↓	0.393	0.283 ↑	0.329 ↑	0.359	0.403 ↑
w/o Amplitude refinement	0.450 ↑	0.443 ↑	0.377 ↑	0.401 ↑	0.385	0.393	0.280 ↑	0.325 ↓	0.363 ↑	0.405 ↑
w/o Phase correction	0.454 ↑	0.442 ↑	0.376 ↑	0.401 ↑	0.384 ↓	0.393	0.279	0.326	0.361 ↑	0.403 ↑
w/o Spectral branch	0.452 ↑	0.438	0.376 ↑	0.401 ↑	0.389 ↑	0.394 ↑	0.283 ↑	0.328 ↑	0.359	0.402
w/o Residual MLP	0.442 ↓	0.437 ↓	0.384 ↑	0.408 ↑	0.392 ↑	0.395 ↑	0.281 ↑	0.326	0.357 ↓	0.402
w/o EDM blocks	0.456 ↑	0.441 ↑	0.379 ↑	0.401 ↑	0.390 ↑	0.395 ↑	0.281 ↑	0.324 ↓	0.364 ↑	0.405 ↑
w/o Signal-adaptive gate	0.447 ↑	0.439 ↑	0.372 ↑	0.398 ↑	0.384 ↓	0.395 ↑	0.278 ↓	0.325 ↓	0.359	0.402
w/o Trend expert	0.445 ↑	0.441 ↑	0.389 ↑	0.413 ↑	0.386 ↑	0.395 ↑	0.279	0.325 ↓	0.364 ↑	0.406 ↑
w/o Learned router	0.455 ↑	0.444 ↑	0.373 ↑	0.400 ↑	0.382 ↓	0.393	0.281 ↑	0.326	0.359	0.403 ↑
w/o KS alignment loss	0.453 ↑	0.441 ↑	0.377 ↑	0.400 ↑	0.385	0.393	0.279	0.326	0.359	0.402
w/o Spectral KL loss	0.447 ↑	0.439 ↑	0.370	0.396 ↓	0.380 ↓	0.393	0.280 ↑	0.326	0.356 ↓	0.400 ↓
w/o Adaptive TDT loss	0.453 ↑	0.446 ↑	0.375 ↑	0.400 ↑	0.382 ↓	0.395 ↑	0.282 ↑	0.327 ↑	0.356 ↓	0.401 ↓
STAR (full model)	**0.443**	**0.438**	**0.370**	**0.397**	**0.385**	**0.393**	**0.279**	**0.326**	**0.359**	**0.402**

TABLE V

EFFICIENCY PROFILE OF STAR ACROSS PREDICTION HORIZONS

Horizon	Params	MACs	Memory (MB)	Latency (ms)
96	52.7K	217M	31.0	0.73
192	74.5K	337M	41.9	0.83
336	107.2K	516M	58.7	1.12
720	194.3K	995M	103.0	1.49

emerges without dataset labels. STAR is evaluated on the ETT family and Exchange, and we frame our conclusions within this evaluated scope. Broader validation on higher-dimensional and more diverse benchmarks, such as Weather, Electricity, Traffic, Solar, and Illness, is a natural next step. Finer-grained routing analysis isolating the horizon prior and the cross-variate spectral co-occupancy signal would further characterize STAR's behavior.

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